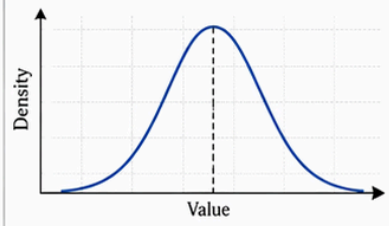
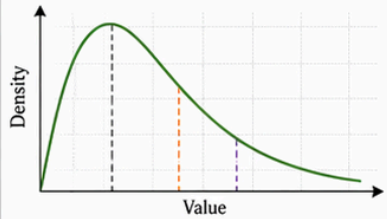
Basic Statistics: Shapes and Distributions

1 Symmetrical Distribution



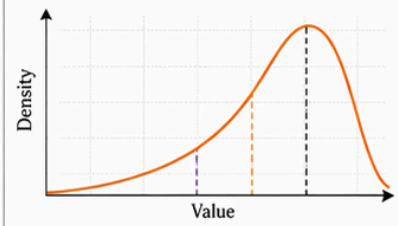
Mean = Median = Mode

2 Positive Skew (Right Skew)



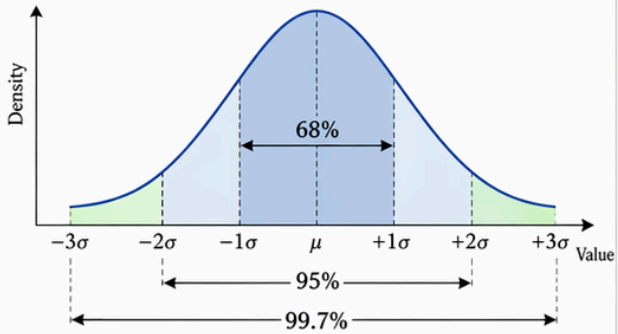
Mean > Median

3 Negative Skew (Left Skew)

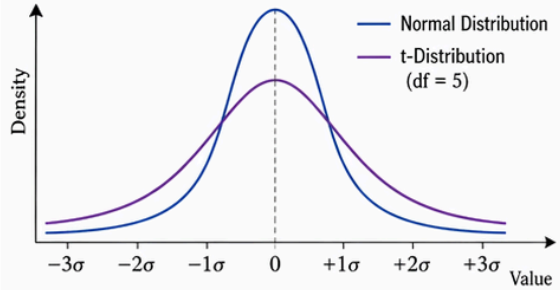


Mean < Median

4 Normal Distribution

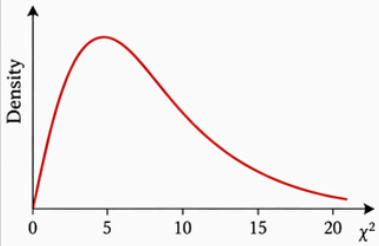


5 t-Distribution vs Normal



Used for small samples

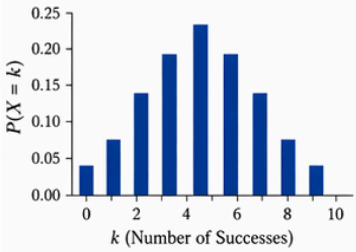
6 Chi-Square Distribution



Used for categorical data

7 Binomial Distribution

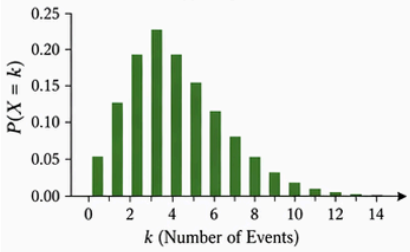
$n = 10, p = 0.5$



Fixed number of trials

8 Poisson Distribution

$\lambda = 4$



Events in fixed interval

9 Uniform Distribution

Equal chance in a range

Key Formulas

Mean

$$\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$$

Binomial

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Poisson

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Uniform

$$f(x) = \frac{1}{b-a}$$

for $a \le x \le b$

Quick memory: Right skew → Mean > Median, Left skew → Mean < Median.

Statistics Notes

What is Statistics?

Statistics is used to **analyze, summarize, and understand data**.
It also helps in **decision making**.

Example:

If we have marks of 100 students, statistics helps us find:

- average marks
- highest and lowest marks
- how marks are spread
- whether performance is improving or not

Types of Statistics

1. Descriptive Statistics

Descriptive statistics is used to **summarize the data we already have**.

It helps us understand data using:

- mean
- median
- mode
- variance
- standard deviation
- shape of distribution

Example:

Marks = 40, 50, 60, 70, 80
Using descriptive statistics, we can find:

- average marks
- middle marks
- whether data is balanced or skewed

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2. Inferential Statistics

Inferential statistics uses **sample data** to make predictions or conclusions about a larger group.

It is often used for:

- prediction
- regression analysis
- hypothesis testing

Example:

If we take data from 100 customers, we can try to predict the behavior of all customers.

Terminology

1. Population

Population means the **entire data set** or whole group we want to study.

Example:

All students in a college.

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2. Sample

Sample means a **small part of the population**.

Example:

100 students selected from the whole college.

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3. Variable

A variable is a **characteristic** that can change from one person or item to another.

Examples:

- age
- gender
- salary
- marks

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4. Data

Data is the information we collect.

- Singular = **datum**
- Plural = **data**

In practice, we usually say **data** only.

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5. Experiment

Experiment means the **process of collecting data**.

Example:

A randomized control trial in medical studies.

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6. Parameter

A parameter is a value that describes the population.

Examples:

- population mean
- population variance

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7. Statistic

A statistic is a numerical value calculated from a sample.

Example:

sample mean, sample standard deviation

Types of Data

1. Qualitative Data

Qualitative data describes qualities or categories.
It is **non-numerical**.

Example:

- color
- gender
- city name
- product type

Types of Qualitative Data

a. Nominal Data

Nominal data has categories, but they **cannot be ordered**.

Examples:

- gender
- blood group
- ethnicity
- seasons

There is no ranking here.

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b. Ordinal Data

Ordinal data can be **ordered**, but the gap between values is not exactly known.

Examples:

- ratings: poor, average, good, excellent
- class rank
- depression scale
- olympic medals

Order exists, but difference is not exact.

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2. Quantitative Data

Quantitative data is **numerical data**.

Example:

- marks
- salary
- age
- height

Types of Quantitative Data

a. Discrete Data

Discrete data is countable, usually whole numbers.

Examples:

- number of children
- number of students
- number of cars
- number of limbs

It is usually finite and integer-based.

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b. Continuous Data

Continuous data can take any value in a range, including decimals.

Examples:

- height
- weight
- distance
- temperature
- time

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3. Interval Data

Interval data has equal intervals, but it does **not have a true zero**.

Examples:

- temperature in Celsius
- temperature in Fahrenheit

Zero does not mean absence.

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4. Ratio Data

Ratio data has equal intervals and a **true zero**.

Examples:

- weight
- height
- age
- income
- distance

Zero means none is present.

Measures of Central Tendency

These help us find the **center** of the data.

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1. Mean

Mean is the average of all values.

Formula:

Mean=Sum of all valuesNumber of values\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}Mean=Number of valuesSum of all values

Example:

Data = 1, 2, 3, 5, 6

Mean=1+2+3+5+65=175=3.4\text{Mean} = \frac{1+2+3+5+6}{5} = \frac{17}{5} = 3.4Mean=51+2+3+5+6=517=3.4

Note:

Mean gets affected by outliers.

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2. Median

Median is the middle value of the ordered data.

Example 1: Odd number of values

Data = 1, 2, 3, 4, 5
Median = 3

Example 2: Even number of values

Data = 1, 2, 3, 4, 5, 6

Median=3+42=3.5\text{Median} = \frac{3+4}{2} = 3.5Median=23+4=3.5

Note:

Median is less affected by outliers.

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3. Mode

Mode is the value that repeats the most.

Example:

Data = 1, 2, 3, 4, 2, 5, 2, 5, 2
Mode = 2

Measures of Dispersion

These tell us how much the data is spread.

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1. Range

Range shows the difference between highest and lowest values.

Formula:

Range=Maximum−Minimum\text{Range} = \text{Maximum} - \text{Minimum}Range=Maximum−Minimum

Example:

Data = 10, 20, 30, 40, 50

Range=50−10=40\text{Range} = 50 - 10 = 40Range=50−10=40

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2. Variance

Variance tells us how far the data points are from the mean.

Simple idea:

- find mean
- subtract mean from each value
- square the results

- find average of those squared values

Formula:

Variance=∑(x−x¯)2n\text{Variance} = \frac{\sum (x - \bar{x})^2}{n}Variance=n∑(x−x¯)2

Example:

If the values are far from the mean, variance will be high.

Use:

Variance helps us understand spread and can help identify outliers.

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3. Standard Deviation

Standard deviation is the square root of variance.

Formula:

Standard Deviation=Variance\text{Standard Deviation} = \sqrt{\text{Variance}}Standard Deviation=Variance

Meaning:

- low standard deviation = data is close to mean
- high standard deviation = data is more spread out

Concept of Distribution

Distribution tells us how the data values are spread.

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1. Symmetric Distribution

In symmetric distribution, the left and right sides are balanced.

Relationship:

Mean=Median=Mode\text{Mean} = \text{Median} = \text{Mode}Mean=Median=Mode

Example:

Bell curve or normal distribution.

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2. Asymmetric Distribution

In asymmetric distribution, data is not balanced.
This is called **skewness**.

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a. Right Skew / Positive Skew

In right skew, the tail is longer on the right side.

Relationship:

Mean>Median\text{Mean} > \text{Median}Mean>Median

Examples:

- household income
- exam scores in some cases

Reason: a few very high values pull the mean to the right.

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b. Left Skew / Negative Skew

In left skew, the tail is longer on the left side.

Relationship:

Mean<Median\text{Mean} < \text{Median}Mean<Median

Example:

A company where most employees have high salary, but a few very low salaries pull the average down.

Probability Density Function

A probability density function shows the likelihood of a **continuous random variable** falling within a range.

It does not usually give probability at one exact point, but over an interval.

Probability Distributions

1. Normal Distribution

Normal distribution is also called **Gaussian distribution**.
It looks like a bell-shaped curve.

Main properties:

- symmetric
- mean = median = mode

Empirical Rule:

- 68% of data lies within 1 standard deviation
- 95% of data lies within 2 standard deviations
- 99.7% of data lies within 3 standard deviations

So only about **0.3%** data is outside 3 standard deviations and can be treated as outliers.

Example:

- height of people
- weight
- exam marks in many cases

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2. Bernoulli Distribution

Bernoulli distribution is for **one experiment with only two outcomes**.

Examples:

- covid test = positive or negative
- diabetes = yes or no
- coin toss = head or tail

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3. Binomial Distribution

Binomial distribution is a combination of many independent Bernoulli trials.

Examples:

- 10 covid tests
- number of students who pass out of 20
- number of heads in 5 coin tosses

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4. Uniform Distribution

Uniform distribution means all outcomes have equal chance.

Examples:

- rolling a dice
- lottery number selection
- random number in a range

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5. Poisson Distribution

Poisson distribution tells the probability of a number of events happening in a fixed interval of time or space.

Examples:

- number of calls received in 1 hour
- number of customers entering a store in 10 minutes

- number of stocks sold in a day

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6. Standard Normal Distribution

This is a special normal distribution where:

- mean = 0
- standard deviation = 1

It is useful when data from different units needs to be compared on the same scale.

Exploratory Data Analysis (EDA)

EDA means studying the data before building a model.

It helps us understand:

- missing values
- outliers
- spread
- relationships between variables
- patterns and trends

Covariance

Covariance tells whether two variables move together.

1. Positive Covariance

If one variable increases, the other also increases.

Example:

As study hours increase, marks may increase.

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2. Negative Covariance

If one variable increases, the other decreases.

Example:

As price increases, demand may decrease.

Collinearity

Collinearity means two features are very similar or strongly related.

Example:

- number of rooms in a house
- number of bedrooms in a house

These two may carry similar information.

Why it matters:

If both features are almost giving the same meaning, the model may overfit or become less stable.

So data engineers or data scientists may:

- keep one feature
- remove one feature
- merge them

Example:

Age and work experience may both predict salary, but they can also be closely related.

Short Final Summary

- **Descriptive statistics** describes existing data
- **Inferential statistics** helps predict for larger population
- **Qualitative data** is descriptive
- **Quantitative data** is numerical
- **Mean, median, mode** show center of data
- **Range, variance, standard deviation** show spread
- **Skewness** shows whether data is tilted left or right
- **Normal distribution** is bell-shaped
- **Bernoulli** is for one yes/no experiment
- **Binomial** is for repeated Bernoulli trials
- **Poisson** counts events in fixed time/space
- **Uniform** means equal chance
- **Covariance** shows relation between two variables
- **Collinearity** means two features are giving similar information